A sharp bound for the least signless Laplacian eigenvalue of triangle-free graphs

Dongxiu Cai[1] Zhengbo Chen[1] Xiao-Dong Zhang[1,2]

Abstract

We prove that every finite simple triangle-free graph G on $n \geq 1$ vertices satisfies $q_{\min}(G) \leq 7n/50$, where $q_{\min}(G)$ is the least eigenvalue of its signless Laplacian matrix $Q_G = D_G + A_G$, with D_G the diagonal matrix of vertex degrees and A_G the adjacency matrix. The constant is sharp, since the Higman–Sims graph and its balanced blow-ups attain equality, as shown by de Lima, Nikiforov and Oliveira (2016). This proves the bound conjectured by Balogh, Clemen, Lidický, Norin and Volec (2023). The proof uses five test column vectors associated with an ordered edge to obtain a matrix Rayleigh inequality. Our use of local Rayleigh inequalities together with flag-algebra semidefinite programming is inspired by Zhou (2026). We use semidefinite programming and rational approximation to find the required positive semidefinite matrices, and complete the proof by exact correction and verification.

2020 Mathematics Subject Classification. Primary 05C50; Secondary 05C35, 90C22.

1 Introduction

Let G be a finite simple graph on $n \geq 1$ vertices, with adjacency matrix A_G. We write $q_{\min}(G)$ for the least eigenvalue of its signless Laplacian

$$Q_G = \mathrm{diag}_{v \in V(G)}(\deg_G(v)) + A_G,$$

where $\deg_G(v) = |\{w \in V(G) : \{v, w\} \in E(G)\}|$ is the degree of v, and $\mathrm{diag}_{z \in S}(b_z)$ denotes the diagonal matrix indexed by a finite set S, with entry b_z at (z, z) and zero at every off-diagonal position. For every real column vector u indexed by $V(G)$,

$$u^{\mathsf{T}} Q_G u = \sum_{\{v,w\} \in E(G)} (u_v + u_w)^2.$$

If G is bipartite, choosing u to be 1 on one part and -1 on the other gives a zero Rayleigh quotient, so $q_{\min}(G) = 0$ by the nonnegativity above. Conversely, if G is connected and $q_{\min}(G) = 0$, a nonzero zero-eigenvector u satisfies $u_v = -u_w$ on every edge; connectedness makes all coordinates nonzero, and their signs give a bipartition. Upper bounds on $q_{\min}(G)$ for triangle-free graphs are related to questions about sparse vertex sets and the number of edges that must be deleted to obtain a bipartite graph [1, 3].

[1] School of Mathematical Sciences, MOE-LSC, SHL-MAC, Shanghai Jiao Tong University, 800 Dongchuan Road, Shanghai 200240, P.R. China. Emails: diudiutse@sjtu.edu.cn (Dongxiu Cai) and czb911@sjtu.edu.cn (Zhengbo Chen).

[2] Corresponding author: Xiao-Dong Zhang. Email: xiaodong@sjtu.edu.cn.

Brandt [3] conjectured that every regular triangle-free graph of order n satisfies $q_{\min}(G) \le 4n/25$. For a d-regular graph, the degree matrix is dI, and therefore

$$q_{\min}(G) = d + \lambda_{\min}(A_G) = \lambda_{\max}(A_G) + \lambda_{\min}(A_G).$$

Consequently, Brandt's conjecture can also be formulated as a bound on the sum of the largest and smallest adjacency eigenvalues. Balogh, Clemen, Lidický, Norin and Volec [1] proved the stronger bound

$$q_{\min}(G) \le \frac{15}{94}n$$

for every triangle-free graph. Their argument uses a test column vector associated with an edge to bound the least eigenvalue by a ratio of induced-path counts, and then bounds that ratio by flag algebras.

The Higman–Sims graph has 100 vertices, is 22-regular and triangle-free, and has least adjacency eigenvalue -8. Its least signless Laplacian eigenvalue is therefore 14. In their concluding remarks, Balogh et al. [1] propose that $q_{\min}(G) \le 0.14n$ holds for every triangle-free graph. They also show that their edge test column vector gives the larger coefficient $42/275$ on the Higman–Sims graph. Improving only the subsequent path-count inequality cannot reach the conjectured coefficient using that same spectral reduction.

We prove the proposed sharp bound.

Theorem 1. *For every finite simple triangle-free graph G with $n = |V(G)| \ge 1$,*

$$q_{\min}(G) \le \frac{7}{50}n.$$

The following known examples show that the bound in Theorem 1 is sharp.

Theorem 2 ([4])**.** *Let K be the Higman–Sims graph. For every integer $t \ge 1$, let $K[t]$ be obtained by replacing each vertex of K by t independent vertices and each edge of K by the complete bipartite graph between the corresponding sets of copies. Then $K[t]$ is triangle-free, has $100t$ vertices, and satisfies*

$$q_{\min}(K[t]) = 14t = \frac{7}{50}|V(K[t])|.$$

Hence $7/50$ is the smallest possible universal coefficient. In particular, the same sharp coefficient bounds $\lambda_{\max}(A_G) + \lambda_{\min}(A_G)$ for regular triangle-free graphs.

The spectral part of the proof uses five test column vectors for each ordered edge. Combining their Rayleigh inequalities produces two real symmetric 5×5 matrices: one records the averaged inner products of the test columns, and the other records their averaged signless Laplacian forms after subtracting the proposed bound. We construct a fixed positive definite matrix whose trace product with the second matrix is nonpositive. Its trace product with the first matrix is strictly positive when the graph has an edge. The matrix Rayleigh inequality then yields the desired eigenvalue bound. One of the test column vectors is obtained by applying a quadratic polynomial in the normalized adjacency matrix to the difference of two adjacency column vectors. This polynomial vanishes at the nonprincipal normalized adjacency eigenvalues of the equality example. The other column vectors also incorporate differences and local variation of vertex degrees.

To prove the trace inequality, we expand the matrix entries into rational linear combinations of graph homomorphism densities. Each term uses at most eight sampled positions. We then compare coefficients on the complete list of 410 triangle-free graphs of order eight. The comparison expresses the required trace as the negative of a sum of nonnegative quadratic forms in conditional flag

probabilities. These quadratic forms use explicit integer matrices whose positivity is verified by strict diagonal dominance. The remaining assertions are 410 exact finite identities. The accompanying data and programs specify the integer entries and check these identities using exact arithmetic.

The use of conditional flag probabilities follows the averaging principle of flag algebras [6]. Our use of local Rayleigh inequalities together with flag-algebra semidefinite programming is inspired by Zhou's work on $\{C_3, C_5\}$-free graphs. Zhou first averages local Rayleigh inequalities [8, Lemma 2.1, pp. 4–5], and then searches jointly for positive semidefinite matrices for the local Rayleigh expressions and ordinary flag squares, followed by rationalization and exact coefficient verification [8, Section 2.3, p. 7]. This is the method we draw on; the five test column vectors associated with an ordered edge and their 5×5 matrix constraint are constructed in the present paper. Here the eight positions are sampled independently with replacement from the original graph. Their sampled values may coincide, while the positions remain distinct vertices of the sampled graph. The resulting identities are exact for every $n \geq 1$, so an eight-vertex calculation proves the bound for all graph orders. We prove the required sampling identities directly.

Section 2 gives the graph and matrix conventions, defines the homomorphism and flag densities used in the proof, and establishes the finite sampling and positivity lemmas. Section 3 constructs the five test column vectors, derives the matrix Rayleigh inequality, specifies the finite integer identities, and proves the bound.

The programs and exact data used in the proof are available at
https://github.com/Diudiutse/triangle-free-signless-laplacian.

2 Preliminary

Graphs and labels. A finite simple graph B is a pair $(V(B), E(B))$, where $V(B)$ is a finite set and

$$E(B) \subseteq \{\{u, v\} : u, v \in V(B),\ u \neq v\}.$$

Edges are unordered two-element sets, so this convention excludes loops and multiple edges, and its adjacency matrix is

$$(A_B)_{uv} = \begin{cases} 1, & \{u, v\} \in E(B), \\ 0, & \{u, v\} \notin E(B), \end{cases} \qquad (u, v \in V(B)).$$

For $S \subseteq V(B)$, the induced subgraph is defined by

$$B[S] = (S,\ E(B) \cap \{\{u, v\} : u, v \in S,\ u \neq v\}),$$

so it retains exactly the edges of B with both endpoints in S. A graph B is triangle-free precisely when

$$\{\{u, v\}, \{u, w\}, \{v, w\}\} \nsubseteq E(B) \quad \text{for all distinct } u, v, w \in V(B).$$

For an integer $s \geq 0$, write $[s] = \{1, \ldots, s\}$, with $[0] = \varnothing$; a graph on $[s]$ is a concrete graph whose vertex names are these integers, so two such graphs are equal exactly when their edge sets are equal. For a map $f : S \to T$ and a subset $R \subseteq S$, write $f(R) = \{f(r) : r \in R\}$ for the image of R and $f|_R : R \to T$, $r \mapsto f(r)$, for the restriction of f to R. A graph isomorphism from B to B' is a bijection preserving both adjacency and nonadjacency, and the set of all such isomorphisms is

$$\mathrm{Iso}(B, B') = \left\{ \phi : V(B) \xrightarrow{\text{bij}} V(B') : (A_B)_{uv} = (A_{B'})_{\phi(u)\phi(v)} \text{ for all } u, v \in V(B) \right\}. \tag{1}$$

We write $B \cong B'$ when $\mathrm{Iso}(B, B') \neq \varnothing$, and this is weaker than equality of graphs on $[s]$, because an isomorphism may permute all vertex names.

Graph maps. For finite simple graphs J, B, define the set of graph homomorphisms by

$$\mathrm{Hom}(J,B) = \{\phi : V(J) \longrightarrow V(B) : \{\phi(u),\phi(v)\} \in E(B) \text{ for every } \{u,v\} \in E(J)\}. \tag{2}$$

A homomorphism may identify nonadjacent vertices of J, but it cannot identify the endpoints of an edge, because an edge cannot be sent to a loop in a simple graph. The injective homomorphisms and the induced embeddings are, respectively,

$$\begin{aligned}\mathrm{Emb}(J,B) &= \{\phi \in \mathrm{Hom}(J,B) : \phi \text{ is injective}\},\\ \mathrm{Emb}_{\mathrm{ind}}(J,B) &= \left\{\phi : V(J) \hookrightarrow V(B) : (A_J)_{uv} = (A_B)_{\phi(u)\phi(v)} \text{ for every } u,v \in V(J)\right\}.\end{aligned} \tag{3}$$

In particular, an induced embedding preserves nonedges as well as edges, whereas an injective homomorphism need only preserve edges. For $n \geq s \geq 0$, write

$$(n)_s = n(n-1)\cdots(n-s+1) = \frac{n!}{(n-s)!}, \qquad (n)_0 = 1,$$

which is the number of injections from an s-element set into an n-element set.

Homomorphism densities. For a graph B of order $n \geq 1$ and a graph J of order s, define

$$t(J,B) = \frac{|\mathrm{Hom}(J,B)|}{n^s}, \qquad t_{\mathrm{inj}}(J,B) = \frac{|\mathrm{Emb}(J,B)|}{(n)_s} \quad (n \geq s). \tag{4}$$

Writing $V(J) = [s]$ converts the first count to the explicit sum

$$t(J,B) = \frac{1}{n^s} \sum_{Y_1,\ldots,Y_s \in V(B)} \prod_{\{a,b\} \in E(J)} (A_B)_{Y_a Y_b}. \tag{5}$$

For a graph J on $[s]$ and $n \geq s$, the ordinary induced-subgraph density is

$$p(J,B) = \frac{|\{S \subseteq V(B) : |S| = s,\ B[S] \cong J\}|}{\binom{n}{s}}. \tag{6}$$

Lemma 1. *If $|V(J)| = s \leq n = |V(B)|$, then*

$$p(J,B) = \frac{s!}{|\mathrm{Iso}(J,J)|} \frac{|\mathrm{Emb}_{\mathrm{ind}}(J,B)|}{(n)_s}.$$

Proof. For each $S \subseteq V(B)$ with $|S| = s$ and $B[S] \cong J$, choose $\psi \in \mathrm{Iso}(J, B[S])$. The map $\phi \mapsto \psi^{-1} \circ \phi$ is a bijection from $\mathrm{Iso}(J, B[S])$ to $\mathrm{Iso}(J,J)$, so

$$|\mathrm{Emb}_{\mathrm{ind}}(J,B)| = |\mathrm{Iso}(J,J)|\,|\{S \subseteq V(B) : |S| = s,\ B[S] \cong J\}|.$$

Dividing by $(n)_s = s!\binom{n}{s}$ gives the stated formula. □

Sampling on positions. For a nonempty graph B and a tuple $X = (X_1,\ldots,X_h) \in V(B)^h$, define the graph on the tuple positions by

$$H_X = ([h],\ \{\{a,b\} : 1 \leq a < b \leq h,\ \{X_a, X_b\} \in E(B)\}). \tag{7}$$

For independent uniform vertex samples, the expectation and probability of any function or event are the explicit finite quantities

$$\mathbb{E}_X f(X) = \frac{1}{|V(B)|^h} \sum_{X \in V(B)^h} f(X), \qquad \Pr(X \in \mathcal{A}) = \frac{|\mathcal{A}|}{|V(B)|^h} \quad (\mathcal{A} \subseteq V(B)^h).$$

Sampling with replacement includes all tuples in $V(B)^h$, whereas sampling without replacement restricts to tuples with pairwise distinct entries and divides by $(|V(B)|)_h$. The ambient graph B is fixed whenever H_X is used, and for $n = |V(B)| \geq h$ the two sampling conventions are related by

$$\Pr(H_X \cong J \mid X_1, \ldots, X_h \text{ are pairwise distinct}) = p(J, B),$$

because each h-element subset admits exactly $h!$ orderings. Let X and Y both be uniformly distributed on $V(B)^h$. For every permutation ρ of $[h]$, the coordinate rearrangement $X \mapsto (X_{\rho(1)}, \ldots, X_{\rho(h)})$ is a bijection of $V(B)^h$, and therefore

$$\mathbb{E}_X f(X_{\rho(1)}, \ldots, X_{\rho(h)}) = \frac{1}{|V(B)|^h} \sum_{Y \in V(B)^h} f(Y) = \mathbb{E}_Y f(Y). \tag{8}$$

Finite graph catalogs. The collection of all triangle-free graphs on the fixed set $[s]$ is

$$\mathcal{G}_s = \left\{ ([s], E) : \begin{array}{l} E \subseteq \{\{a, b\} : 1 \leq a < b \leq s\}, \\ ([s], E) \text{ is triangle-free} \end{array} \right\}. \tag{9}$$

For each $s = 0, \ldots, 8$, the finite catalog $\mathcal{T}_s \subseteq \mathcal{G}_s$ is required to satisfy

$$|\{H \in \mathcal{T}_s : B \cong H\}| = 1 \qquad \text{for every } B \in \mathcal{G}_s, \tag{10}$$

so the catalog contains one concrete edge set on $[s]$ from each ordinary isomorphism class. The vertex names $1, \ldots, s$ have already been specified in (9); the choice in (10) concerns which edge set represents a class, and does not choose a new set of vertex labels. If the ambient graph B is triangle-free, then H_X is triangle-free: three pairwise adjacent positions would have three distinct values, since adjacent positions cannot have equal values, and those three values would form a triangle in B. Thus, for $0 \leq h \leq 8$, independent h-tuples from any nonempty triangle-free B give a partition

$$V(B)^h = \bigsqcup_{H \in \mathcal{T}_h} \{X \in V(B)^h : H_X \cong H\}, \qquad \sum_{H \in \mathcal{T}_h} \Pr(H_X \cong H) = 1. \tag{11}$$

The following finite sampling identity uses the same averaging idea as [5, proof of Proposition 11.32]. We give a direct proof using independent uniform samples from $V(B)$.

Lemma 2. *Let B be a nonempty triangle-free graph, let $1 \leq h \leq 8$, and let J be a simple graph of order $s \leq h$, including $s = 0$. For independent uniform $X_1, \ldots, X_h \in V(B)$,*

$$t(J, B) = \mathbb{E}_X t_{\mathrm{inj}}(J, H_X) = \sum_{H \in \mathcal{T}_h} t_{\mathrm{inj}}(J, H) \Pr(H_X \cong H). \tag{12}$$

Proof. Fix an injection $\phi : V(J) \hookrightarrow [h]$; its images are different positions, so the values $(X_{\phi(a)})_{a \in V(J)}$ are independent uniform samples, although those values may coincide. Using (7) and (5), this gives

$$\mathbb{E}_X \prod_{\{a,b\} \in E(J)} (A_{H_X})_{\phi(a)\phi(b)} = \mathbb{E}_X \prod_{\{a,b\} \in E(J)} (A_B)_{X_{\phi(a)} X_{\phi(b)}} = t(J, B).$$

There are $(h)_s$ possible injections, so averaging their equal expectations gives

$$\begin{aligned}\mathbb{E}_X t_{\mathrm{inj}}(J, H_X) &= \frac{1}{(h)_s} \sum_{\phi: V(J) \hookrightarrow [h]} \mathbb{E}_X \prod_{\{a,b\} \in E(J)} (A_{H_X})_{\phi(a)\phi(b)} \\ &= \frac{1}{(h)_s} (h)_s \, t(J, B) = t(J, B).\end{aligned}$$

If $\eta : H \to H'$ is an isomorphism, then $\phi \mapsto \eta \circ \phi$ is a bijection from $\mathrm{Emb}(J, H)$ to $\mathrm{Emb}(J, H')$, so $t_{\mathrm{inj}}(J, H) = t_{\mathrm{inj}}(J, H')$. This invariance makes $t_{\mathrm{inj}}(J, H_X)$ constant on each part of (11), and summing the expectation over that partition proves the second equality in (12). □

Types and ordered roots. A type of order k is a specified triangle-free graph $\sigma = ([k], E(\sigma))$, with $0 \le k \le 6$, and the proof uses each concrete $\sigma \in \mathcal{T}_k$ once. A σ-rooted graph is a pair (B, θ), where B is triangle-free and

$$\theta \in \mathrm{Emb}_{\mathrm{ind}}(\sigma, B), \tag{13}$$

so the ordered roots are the vertices $\theta(1), \dots, \theta(k)$ and their adjacency matrix is exactly A_σ. Two σ-rooted graphs are isomorphic exactly when

$$(B, \theta) \cong_\sigma (B', \theta') \quad \Longleftrightarrow \quad \{\phi \in \mathrm{Iso}(B, B') : \phi(\theta(i)) = \theta'(i) \ (i \in [k])\} \neq \varnothing. \tag{14}$$

A σ-flag is an isomorphism class under (14), and a flag of order ℓ has ℓ vertices, of which exactly k are roots and $\ell - k$ are free vertices. Every rooted graph (B, θ) of order ℓ can be represented on $[\ell]$: choose a bijection $\lambda : [\ell] \to V(B)$ with $\lambda(i) = \theta(i)$ for $i \in [k]$ and use

$$([\ell], \{\{a, b\} : 1 \le a < b \le \ell, \ \{\lambda(a), \lambda(b)\} \in E(B)\}).$$

If λ' is another such bijection, then $(\lambda')^{-1} \circ \lambda$ fixes every $i \in [k]$ and is an isomorphism between the two graphs on $[\ell]$, so the resulting rooted isomorphism class does not depend on the ordering of the free vertices. For $\ell \ge k$, the flag catalog $\mathcal{F}_{\sigma,\ell}$ consists of graphs satisfying

$$\mathcal{F}_{\sigma,\ell} \subseteq \{P \in \mathcal{G}_\ell : P[[k]] = \sigma\}, \qquad |\{P \in \mathcal{F}_{\sigma,\ell} : (B, \iota) \cong_\sigma (P, \iota)\}| = 1 \tag{15}$$

for every $B \in \mathcal{G}_\ell$ with $B[[k]] = \sigma$, where $\iota : [k] \hookrightarrow [\ell]$ is the inclusion $\iota(i) = i$. For a catalog graph P, we also write P for the rooted pair (P, ι), so $(B, \theta) \cong_\sigma P$ is always the explicit isomorphism condition in (14). For the empty type $k = 0$, there is a unique empty root map, (14) reduces to ordinary graph isomorphism, and flag densities reduce to ordinary induced-subgraph densities.

Flag densities. If (B, θ) is σ-rooted, $|V(B)| = h$, and $P \in \mathcal{F}_{\sigma,\ell}$ with $k \le \ell \le h$, define

$$p(P; (B, \theta)) = \frac{|\{S \subseteq V(B) \setminus \theta([k]) : |S| = \ell - k, \ (B[\theta([k]) \cup S], \theta) \cong_\sigma P\}|}{\binom{h-k}{\ell-k}}. \tag{16}$$

The induced graph in (16) retains every edge and nonedge among its roots and its free set, and θ remains the same ordered root map after restricting the graph. For $P \in \mathcal{F}_{\sigma,\ell}$, $P' \in \mathcal{F}_{\sigma,\ell'}$, and $h \ge \ell + \ell' - k$, define the joint density by

$$p(P, P'; (B, \theta)) = \frac{\left| \left\{ (S, S') : \begin{array}{l} S, S' \subseteq V(B) \setminus \theta([k]), \quad S \cap S' = \varnothing, \\ |S| = \ell - k, \quad |S'| = \ell' - k, \\ (B[\theta([k]) \cup S], \theta) \cong_\sigma P, \\ (B[\theta([k]) \cup S'], \theta) \cong_\sigma P' \end{array} \right\} \right|}{\binom{h-k}{\ell-k} \binom{h-\ell}{\ell'-k}}. \tag{17}$$

The denominator first chooses the free set for P and then chooses the free set for P' from the remaining vertices, so the numerator counts an ordered pair of disjoint free sets and tests no edges between those sets. Every eligible free set, or pair of free sets, belongs to exactly one catalog flag, or pair of catalog flags, so

$$\sum_{P\in\mathcal{F}_{\sigma,\ell}} p(P;(B,\theta)) = 1, \qquad \sum_{\substack{P\in\mathcal{F}_{\sigma,\ell}\\ P'\in\mathcal{F}_{\sigma,\ell'}}} p(P,P';(B,\theta)) = 1.$$

Lemma 3 ([6, Lemma 2.2])**.** *Let (B,θ) be a σ-rooted graph of order h, and let R range over $\mathcal{F}_{\sigma,d}$. For $k \le \ell \le d \le h$ and $P \in \mathcal{F}_{\sigma,\ell}$,*

$$p(P;(B,\theta)) = \sum_R p(P;R)\, p(R;(B,\theta)).$$

If $P' \in \mathcal{F}_{\sigma,\ell'}$ and $\ell + \ell' - k \le d \le h$, then

$$p(P,P';(B,\theta)) = \sum_R p(P,P';R)\, p(R;(B,\theta)).$$

The flag product is defined by these joint-density coefficients [6, equation (5) and Lemma 2.4]: at any order $d \ge \ell + \ell' - k$, the product of P, P' is represented by

$$\sum_{R\in\mathcal{F}_{\sigma,d}} p(P,P';R)\, R,$$

and Lemma 3 shows how these coefficients behave when evaluated on a larger rooted graph. For a formal linear combination $\sum_P c_P P$ with all P of order ℓ, the coefficient of R in its square is

$$\sum_{P,P'\in\mathcal{F}_{\sigma,\ell}} c_P c_{P'}\, p(P,P';R),$$

obtained by distributing the two finite sums.

Integer ordering counts. For each type σ of order k, we use flags of the order

$$\ell = \left\lfloor \frac{8+k}{2} \right\rfloor, \qquad 2\ell - k \le 8, \qquad \mathcal{F}_\sigma = \mathcal{F}_{\sigma,\ell},$$

with ℓ specifying the number of vertices of a flag whenever this type is being considered. For any triangle-free graph H with eight vertices and $P, P' \in \mathcal{F}_\sigma$, define the integer matrix entry

$$C_H^\sigma(P,P') = \left| \left\{ \rho : \begin{array}{l} \rho : [8] \xrightarrow{\text{bij}} V(H), \\ (A_H)_{\rho(i)\rho(j)} = (A_\sigma)_{ij} \quad (1 \le i < j \le k), \\ (H[\rho([\ell])], \rho|_{[k]}) \cong_\sigma P, \\ (H[\rho([k] \cup \{\ell+1,\dots,2\ell-k\})], \rho|_{[k]}) \cong_\sigma P' \end{array} \right\} \right|. \tag{18}$$

The two induced graphs in (18) use the same roots and the disjoint free-position sets

$$\{k+1,\dots,\ell\}, \qquad \{\ell+1,\dots,2\ell-k\}.$$

Clearly, C_H^σ is symmetric, that is, $C_H^\sigma(P,P') = C_H^\sigma(P',P)$ for all $P, P' \in \mathcal{F}_\sigma$.

Lemma 4. *For every triangle-free graph H of order eight,*

$$\frac{C_H^\sigma(P,P')}{8!} = \frac{1}{(8)_k} \sum_{\theta \in \mathrm{Emb}_{\mathrm{ind}}(\sigma,H)} p(P,P';(H,\theta)). \tag{19}$$

Proof. We express the root-averaging convention of [6, Section 2.2] in terms of the ordering count (18). Fix $\theta \in \mathrm{Emb}_{\mathrm{ind}}(\sigma,H)$ and an ordered pair of disjoint sets $S,S' \subseteq V(H)\setminus\theta([k])$, each of size $\ell-k$. There are exactly

$$(\ell-k)!^2(8-2\ell+k)!$$

bijections $\rho:[8]\to V(H)$ satisfying

$$\rho|_{[k]} = \theta, \qquad \rho(\{k+1,\dots,\ell\}) = S, \qquad \rho(\{\ell+1,\dots,2\ell-k\}) = S',$$

because the two free groups and the unused positions can be ordered independently. Thus (18) and (17) give

$$\begin{aligned}
\frac{C_H^\sigma(P,P')}{8!} &= \frac{(\ell-k)!^2(8-2\ell+k)!}{8!}\binom{8-k}{\ell-k}\binom{8-\ell}{\ell-k} \sum_{\theta \in \mathrm{Emb}_{\mathrm{ind}}(\sigma,H)} p(P,P';(H,\theta)) \\
&= \frac{(8-k)!}{8!} \sum_{\theta \in \mathrm{Emb}_{\mathrm{ind}}(\sigma,H)} p(P,P';(H,\theta)) \\
&= \frac{1}{(8)_k} \sum_{\theta \in \mathrm{Emb}_{\mathrm{ind}}(\sigma,H)} p(P,P';(H,\theta)).
\end{aligned}$$

This proves (19). □

Summing (19) over all $P,P' \in \mathcal{F}_\sigma$ gives the useful normalization

$$\frac{1}{8!}\sum_{P,P'} C_H^\sigma(P,P') = \frac{|\,\mathrm{Emb}_{\mathrm{ind}}(\sigma,H)|}{(8)_k}.$$

Conditional flag probabilities. Now fix a nonempty triangle-free graph B of order n, a tuple $r=(r_1,\dots,r_k)\in V(B)^k$, and a flag $P\in\mathcal{F}_\sigma$, and let H_r be the position graph defined by (7). The conditional probability used on B is defined by

$$\Pr(P \mid r) = \begin{cases} \dfrac{\left|\left\{Y \in V(B)^{\ell-k} : (H_{(r,Y)},\iota) \cong_\sigma P\right\}\right|}{n^{\ell-k}}, & H_r = \sigma, \\ 0, & H_r \neq \sigma, \end{cases} \tag{20}$$

where $(r,Y)=(r_1,\dots,r_k,Y_1,\dots,Y_{\ell-k})$ and $\iota(i)=i$ maps the roots into the position graph. For $H_r=\sigma$, the sets of tuples Y producing the catalog flags partition $V(B)^{\ell-k}$, so

$$\sum_{P\in\mathcal{F}_\sigma} \Pr(P \mid r) = 1,$$

while every summand is zero when $H_r\neq\sigma$. For $k=0$, the root tuple is the unique empty tuple, H_r is the empty type, and the root average consists of one term with probability one. The joint event for a tuple of eight samples is the set

$$\mathcal{A}_{P,P'}^\sigma = \left\{ X\in V(B)^8 : \begin{array}{l} H_{(X_1,\dots,X_k)} = \sigma, \\ (H_{(X_1,\dots,X_\ell)},\iota) \cong_\sigma P, \\ (H_{(X_1,\dots,X_k,X_{\ell+1},\dots,X_{2\ell-k})},\iota) \cong_\sigma P' \end{array} \right\}. \tag{21}$$

The nonnegativity of averaged flag squares is a standard result [6, Theorem 3.1(a)]. The next lemma records the exact identity needed for the sampling rule (7), including tuples whose values in $V(B)$ repeat. Its proof verifies this identity with the normalization 8! in (18).

Lemma 5. *For every nonempty triangle-free B, independent uniform $X \in V(B)^8$, and $r = (X_1, \dots, X_k)$,*

$$\frac{1}{8!} \sum_{H \in \mathcal{T}_8} \Pr(H_X \cong H) C_H^\sigma(P, P') = \mathbb{E}_r \left[\mathbf{1}_{H_r = \sigma} \Pr(P \mid r) \Pr(P' \mid r) \right]. \tag{22}$$

Consequently, for arbitrary real coordinates $(w_P)_{P \in \mathcal{F}_\sigma}$,

$$\begin{aligned} &\frac{1}{8!} \sum_{H \in \mathcal{T}_8} \Pr(H_X \cong H) \sum_{P, P'} w_P C_H^\sigma(P, P') w_{P'} \\ &\qquad = \mathbb{E}_r \left[\mathbf{1}_{H_r = \sigma} \left(\sum_P w_P \Pr(P \mid r) \right)^2 \right] \geq 0. \end{aligned} \tag{23}$$

In particular,

$$\sum_{H \in \mathcal{T}_8} \Pr(H_X \cong H) C_H^\sigma \quad \textit{is positive semidefinite.} \tag{24}$$

Proof. Write $n = |V(B)|$ and $S_8 = \{\rho : [8] \xrightarrow{\text{bij}} [8]\}$. For $\rho \in S_8$, the two induced graphs in (18) are identified with the position graphs in (21) by the maps

$$\begin{aligned} &[\ell] \longrightarrow \rho([\ell]), && i \longmapsto \rho(i), \\ &[\ell] \longrightarrow \rho([k] \cup \{\ell + 1, \dots, 2\ell - k\}), && i \longmapsto \begin{cases} \rho(i), & i \leq k, \\ \rho(\ell + i - k), & i > k. \end{cases} \end{aligned}$$

Both maps send root i to $\rho(i)$ and preserve adjacency by (7); hence

$$C_{H_X}^\sigma(P, P') = \left| \left\{ \rho \in S_8 : (X_{\rho(1)}, \dots, X_{\rho(8)}) \in \mathcal{A}_{P,P'}^\sigma \right\} \right|.$$

For every $\rho \in S_8$, the maps

$$(X_1, \dots, X_8) \longmapsto (X_{\rho(1)}, \dots, X_{\rho(8)}), \qquad (Y_1, \dots, Y_8) \longmapsto (Y_{\rho^{-1}(1)}, \dots, Y_{\rho^{-1}(8)})$$

are inverse bijections of $V(B)^8$, so

$$\left| \left\{ X \in V(B)^8 : (X_{\rho(1)}, \dots, X_{\rho(8)}) \in \mathcal{A}_{P,P'}^\sigma \right\} \right| = |\mathcal{A}_{P,P'}^\sigma|.$$

Using the partition (11) and the isomorphism invariance of C_H^σ, we obtain

$$\begin{aligned}
&\frac{1}{8!}\sum_{H\in\mathcal{T}_8}\Pr(H_X\cong H)C_H^\sigma(P,P')\\
&=\frac{1}{8!\,n^8}\sum_{H\in\mathcal{T}_8}\sum_{\substack{X\in V(B)^8\\ H_X\cong H}}C_H^\sigma(P,P')\\
&=\frac{1}{8!\,n^8}\sum_{X\in V(B)^8}C_{H_X}^\sigma(P,P')\\
&=\frac{1}{8!\,n^8}\sum_{X\in V(B)^8}\sum_{\rho\in S_8}\mathbf{1}_{\{(X_{\rho(1)},\dots,X_{\rho(8)})\in\mathcal{A}_{P,P'}^\sigma\}}\\
&=\frac{1}{8!\,n^8}\sum_{\rho\in S_8}\left|\left\{X\in V(B)^8:(X_{\rho(1)},\dots,X_{\rho(8)})\in\mathcal{A}_{P,P'}^\sigma\right\}\right|\\
&=\frac{1}{8!\,n^8}\sum_{\rho\in S_8}|\mathcal{A}_{P,P'}^\sigma|=\frac{|\mathcal{A}_{P,P'}^\sigma|}{n^8}.
\end{aligned}$$

For each fixed $r\in V(B)^k$, the definition (21) gives the Cartesian product

$$\begin{aligned}
&\left\{(Y,Z,W):\ \begin{array}{l}Y,Z\in V(B)^{\ell-k},\quad W\in V(B)^{8-2\ell+k},\\ (r,Y,Z,W)\in\mathcal{A}_{P,P'}^\sigma\end{array}\right\}\\
&\quad=\begin{cases}\{Y\in V(B)^{\ell-k}:(H_{(r,Y)},\iota)\cong_\sigma P\}\\ \quad\times\{Z\in V(B)^{\ell-k}:(H_{(r,Z)},\iota)\cong_\sigma P'\} & H_r=\sigma,\\ \quad\times V(B)^{8-2\ell+k},\\ \varnothing, & H_r\neq\sigma.\end{cases}
\end{aligned}$$

Here (r,Y,Z,W) is concatenation in the displayed order; its blocks occupy positions $1,\dots,k$, $k+1,\dots,\ell$, $\ell+1,\dots,2\ell-k$, and $2\ell-k+1,\dots,8$. Taking cardinalities and applying (20) gives

$$\begin{aligned}
\frac{|\mathcal{A}_{P,P'}^\sigma|}{n^8}&=\frac{1}{n^8}\sum_{r\in V(B)^k}\mathbf{1}_{H_r=\sigma}\\
&\qquad\cdot\left|\{Y\in V(B)^{\ell-k}:(H_{(r,Y)},\iota)\cong_\sigma P\}\right|\\
&\qquad\cdot\left|\{Z\in V(B)^{\ell-k}:(H_{(r,Z)},\iota)\cong_\sigma P'\}\right|\ n^{8-2\ell+k}\\
&=\frac{1}{n^8}\sum_{r\in V(B)^k}\mathbf{1}_{H_r=\sigma}\,n^{\ell-k}\Pr(P\mid r)\,n^{\ell-k}\Pr(P'\mid r)\,n^{8-2\ell+k}\\
&=\frac{1}{n^k}\sum_{r\in V(B)^k}\mathbf{1}_{H_r=\sigma}\Pr(P\mid r)\Pr(P'\mid r)\\
&=\mathbb{E}_r\left[\mathbf{1}_{H_r=\sigma}\Pr(P\mid r)\Pr(P'\mid r)\right],
\end{aligned}$$

where $(\ell-k)+(\ell-k)+(8-2\ell+k)=8-k$. This proves (22). Summing (22) with coefficients $w_Pw_{P'}$ gives (23) by linearity of expectation. This finishes the proof of Lemma 5. □

Matrix positivity and traces. A real symmetric matrix B is positive semidefinite when $u^{\mathsf{T}}Bu\geq 0$ for every real column vector u, and positive definite when $u^{\mathsf{T}}Bu>0$ for every nonzero u;

$\operatorname{tr}(B) = \sum_i B_{ii}$ denotes its trace. For real matrices B, C whose products are defined and square, direct summation gives

$$\operatorname{tr}(BC) = \sum_{i,j} B_{ij} C_{ji} = \sum_{j,i} C_{ji} B_{ij} = \operatorname{tr}(CB),$$

and for symmetric B, C this is also $\sum_{i,j} B_{ij} C_{ij}$.

Lemma 6 ([2, Example 2.24])**.** *If B, C are real symmetric positive semidefinite matrices of the same size, then* $\operatorname{tr}(BC) \geq 0$.

Lemma 7. *Let X be a symmetric integer matrix and let $q > 0$ be an integer. If every row satisfies*

$$q + X_{ii} - \sum_{j \neq i} |X_{ij}| > 0,$$

then $qI + X$ is positive definite.

Proof. The matrix $qI + X$ is real symmetric, so all its eigenvalues are real. By Gershgorin's theorem [7, Theorem 1.1], for every eigenvalue λ there is a row i such that

$$|\lambda - (q + X_{ii})| \leq \sum_{j \neq i} |X_{ij}|.$$

Hence

$$\lambda \geq q + X_{ii} - \sum_{j \neq i} |X_{ij}| > 0,$$

so $qI + X$ is positive definite. □

For any real matrix V and any positive semidefinite B, the product VBV^{T} is positive semidefinite because

$$u^{\mathsf{T}} V B V^{\mathsf{T}} u = (V^{\mathsf{T}} u)^{\mathsf{T}} B (V^{\mathsf{T}} u) \geq 0,$$

and it is positive definite when B is positive definite and V is square and invertible.

The graph in the theorem. In the proof, $G = (V, E(G))$ is a finite simple triangle-free graph with $n = |V| \geq 1$, $A = A_G$, and

$$\deg_G(v) = |\{u \in V : \{u, v\} \in E(G)\}| = \sum_{u \in V} A_{vu}.$$

The least eigenvalue of the real symmetric matrix

$$Q_G = \operatorname{diag}_{v \in V}(\deg_G(v)) + A$$

is denoted by $q_{\min}(G)$. The adjacency column vector $A_{:x}$ is indexed by V and has coordinate $(A_{:x})(z) = A_{zx}$, while for a real column vector u indexed by V,

$$\|u\|^2 = u^{\mathsf{T}} u = \sum_{z \in V} u(z)^2,$$

and I denotes the identity matrix of the size required by its displayed product. Subscripts $x, y, z, \ldots$ on $\mathbb{E}$ mean independent uniform samples from V with replacement. In particular, each ordered pair $(x, y) \in V^2$ has probability $1/n^2$. Since $A_{xy} = 1$ when $\{x, y\} \in E(G)$ and $A_{xy} = 0$ otherwise, for every function $f : V^2 \to \mathbb{R}$,

$$\mathbb{E}_{x,y}[A_{xy} f(x, y)] = \frac{1}{n^2} \sum_{(x,y) \in V^2} A_{xy} f(x, y) = \frac{1}{n^2} \sum_{\substack{(x,y) \in V^2 \\ \{x,y\} \in E(G)}} f(x, y).$$

Each edge $\{x, y\}$ contributes both ordered pairs (x, y) and (y, x), and each term still has weight $1/n^2$.

3 Proof of the main theorem

To prove Theorem 1, we first give a sufficient condition in Lemma 8. In order to state this lemma, we introduce the test column vectors and the two averaged matrices used below.

For every ordered pair $(x, y) \in V^2$, define the first test column vector by

$$U_0^{xy} = \left(\frac{A}{n} - \frac{1}{50}I\right)(A_{:x} - A_{:y}), \tag{25}$$

and, for every $z \in V$, matrix multiplication gives

$$\begin{aligned} U_0^{xy}(z) &= \sum_{t\in V}\left(\frac{A}{n} - \frac{1}{50}I\right)_{zt}(A_{:x} - A_{:y})(t) \\ &= \sum_{t\in V}\left(\frac{A_{zt}}{n} - \frac{I_{zt}}{50}\right)(A_{tx} - A_{ty}) \\ &= \frac{1}{n}\sum_{t\in V} A_{zt}(A_{tx} - A_{ty}) - \frac{1}{50}\sum_{t\in V} I_{zt}(A_{tx} - A_{ty}) \\ &= \frac{1}{n}\sum_{t\in V} A_{zt}(A_{tx} - A_{ty}) - \frac{A_{zx} - A_{zy}}{50}, \end{aligned}$$

where $(A_{:x} - A_{:y})(t) = A_{tx} - A_{ty}$ and $I_{zt} = 1$ for $t = z$ and $I_{zt} = 0$ for $t \neq z$. If $A_{xy} = 1$, triangle-freeness gives $A_{xt}A_{yt} = 0$ for every $t \in V$, because a common neighbor of the edge endpoints would form a triangle. Using this vanishing product, symmetry, $A_{xx} = A_{yy} = 0$, $A_{xy} = A_{yx} = 1$, and $A_{zt}^2 = A_{zt}$, the endpoint coordinates are

$$\begin{aligned} U_0^{xy}(x) &= \frac{1}{n}\sum_t (A_{xt}^2 - A_{xt}A_{yt}) - \frac{0-1}{50} = \frac{\deg_G(x)}{n} + \frac{1}{50} > 0, \\ U_0^{xy}(y) &= \frac{1}{n}\sum_t (A_{yt}A_{xt} - A_{yt}^2) - \frac{1-0}{50} = -\frac{\deg_G(y)}{n} - \frac{1}{50} < 0. \end{aligned} \tag{26}$$

In particular, (26) proves that $U_0^{xy} \neq 0$ for every ordered edge. Define the next two column vectors directly from the adjacency column vectors and the degree diagonal entries by

$$\begin{aligned} U_1^{xy} &= \frac{\deg_G(x) - \deg_G(y)}{n}(A_{:x} + A_{:y}), \\ U_2^{xy} &= \operatorname{diag}_{z\in V}\left(\frac{\deg_G(z)}{n} - \frac{\deg_G(x) + \deg_G(y)}{2n}\right)(A_{:x} - A_{:y}). \end{aligned} \tag{27}$$

The fourth column vector is

$$U_3^{xy} = \left(\frac{A^2}{n^2} + \frac{3A}{50n} - \frac{1}{625}I\right)(A_{:x} - A_{:y}). \tag{28}$$

The numerical coefficients in (28) can also be read from the explicit multiplication

$$\begin{aligned} \left(\frac{A}{n} + \frac{2}{25}I\right)U_0^{xy} &= \left(\frac{A}{n} + \frac{2}{25}I\right)\left(\frac{A}{n} - \frac{1}{50}I\right)(A_{:x} - A_{:y}) \\ &= \left(\frac{A^2}{n^2} + \left(\frac{2}{25} - \frac{1}{50}\right)\frac{A}{n} - \frac{2}{25\cdot 50}I\right)(A_{:x} - A_{:y}) \\ &= U_3^{xy}. \end{aligned}$$

The last column vector is

$$U_4^{xy} = \operatorname{diag}_{z\in V}\left(\frac{\deg_G(z)}{n} - \frac{\deg_G(x)+\deg_G(y)}{2n}\right) U_0^{xy} + \frac{(\deg_G(x)-\deg_G(y))^2}{4n^2}(A_{:x} - A_{:y}), \tag{29}$$

and U^{xy} is the $n \times 5$ matrix with these columns $U_0^{xy}, \ldots, U_4^{xy}$ in order.

For $i, j = 0, \ldots, 4$, define

$$\begin{aligned} \mathsf{D}_{ij} &= \mathbb{E}_{x,y}\left[\frac{A_{xy}}{n}(U_i^{xy})^{\mathsf{T}} U_j^{xy}\right], \\ \mathsf{F}_{ij} &= \mathbb{E}_{x,y}\left[\frac{A_{xy}}{n}(U_i^{xy})^{\mathsf{T}}\left(\frac{Q_G}{n} - \frac{7}{50}I\right) U_j^{xy}\right]. \end{aligned} \tag{30}$$

Since Q_G is real symmetric, transposing either quadratic product in (30) exchanges i, j without changing its value, so both D, F are real symmetric 5×5 matrices.

Lemma 8. *If there exists a real symmetric positive definite 5×5 matrix B such that, for every finite simple triangle-free graph G with $n = |V(G)| \geq 1$,*

$$\operatorname{tr}(B\mathsf{F}) \leq 0,$$

where F is the matrix defined from that graph G by (30), then Theorem 1 holds. Here the same matrix B is used for every G.

Proof. Let B be a matrix satisfying the hypotheses, and fix an arbitrary finite simple triangle-free graph G with $n = |V(G)| \geq 1$. If $E(G) = \varnothing$, then $Q_G = 0$ and $q_{\min}(G) = 0 \leq 7n/50$. It therefore suffices to consider $E(G) \neq \varnothing$. Choose an orthonormal eigenbasis $e_1, \ldots, e_n$ of Q_G, with corresponding eigenvalues $\lambda_1, \ldots, \lambda_n$, so every $\lambda_j \geq q_{\min}(G)$. For every column vector $u \in \mathbb{R}^V$,

$$u^{\mathsf{T}}(Q_G - q_{\min}(G)I)u = \sum_{j=1}^{n}(\lambda_j - q_{\min}(G))(e_j^{\mathsf{T}}u)^2 \geq 0,$$

and division by $n > 0$ gives

$$u^{\mathsf{T}}\left(\frac{Q_G}{n}\right)u \geq \frac{q_{\min}(G)}{n}\, u^{\mathsf{T}}u. \tag{31}$$

Fix $u = (u_0, \ldots, u_4)^{\mathsf{T}} \in \mathbb{R}^5$. The columns of U^{xy} give $U^{xy}u = \sum_{i=0}^{4} u_i U_i^{xy}$, so the first definition in

(30) yields the entrywise calculation

$$\begin{aligned}
u^{\mathsf{T}}\mathsf{D}u &= \sum_{i=0}^{4}\sum_{j=0}^{4} u_i \mathsf{D}_{ij} u_j \\
&= \sum_{i=0}^{4}\sum_{j=0}^{4} u_i u_j \mathbb{E}_{x,y}\left[\frac{A_{xy}}{n}(U_i^{xy})^{\mathsf{T}} U_j^{xy}\right] \\
&= \mathbb{E}_{x,y}\left[\frac{A_{xy}}{n}\sum_{i=0}^{4}\sum_{j=0}^{4} u_i u_j (U_i^{xy})^{\mathsf{T}} U_j^{xy}\right] \\
&= \mathbb{E}_{x,y}\left[\frac{A_{xy}}{n}\left(\sum_{i=0}^{4} u_i U_i^{xy}\right)^{\mathsf{T}}\left(\sum_{j=0}^{4} u_j U_j^{xy}\right)\right] \\
&= \mathbb{E}_{x,y}\left[\frac{A_{xy}}{n}(U^{xy}u)^{\mathsf{T}}(U^{xy}u)\right] \\
&= \mathbb{E}_{x,y}\left[\frac{A_{xy}}{n}\|U^{xy}u\|^2\right] \\
&= \mathbb{E}_{x,y}\left[\frac{A_{xy}}{n}\sum_{z\in V}\left(\sum_{i=0}^{4} u_i U_i^{xy}(z)\right)^2\right] \geq 0.
\end{aligned}$$

The last inequality uses $A_{xy}/n \geq 0$ and the nonnegativity of every square; hence D is positive semidefinite. Its diagonal entries are averages of squared norms, and (26) and $E(G) \neq \varnothing$ give

$$\operatorname{tr}(\mathsf{D}) = \sum_{i=0}^{4}\mathbb{E}_{x,y}\left[\frac{A_{xy}}{n}\|U_i^{xy}\|^2\right] \geq \mathbb{E}_{x,y}\left[\frac{A_{xy}}{n}\|U_0^{xy}\|^2\right] > 0. \tag{32}$$

For every $u = (u_0, \ldots, u_4)^{\mathsf{T}} \in \mathbb{R}^5$, substitute both definitions in (30) and use $U^{xy}u = \sum_{i=0}^{4} u_i U_i^{xy}$:

$$\begin{aligned}
&u^{\mathsf{T}}\left[\mathsf{F} - \left(\frac{q_{\min}(G)}{n} - \frac{7}{50}\right)\mathsf{D}\right]u \\
&= \sum_{i=0}^{4}\sum_{j=0}^{4} u_i u_j \left[\mathsf{F}_{ij} - \left(\frac{q_{\min}(G)}{n} - \frac{7}{50}\right)\mathsf{D}_{ij}\right] \\
&= \mathbb{E}_{x,y}\left[\frac{A_{xy}}{n}\sum_{i=0}^{4}\sum_{j=0}^{4} u_i u_j (U_i^{xy})^{\mathsf{T}}\left(\frac{Q_G}{n} - \frac{7}{50}I - \left(\frac{q_{\min}(G)}{n} - \frac{7}{50}\right)I\right)U_j^{xy}\right] \\
&= \mathbb{E}_{x,y}\left[\frac{A_{xy}}{n}\left(\sum_{i=0}^{4} u_i U_i^{xy}\right)^{\mathsf{T}}\left(\frac{Q_G}{n} - \frac{7}{50}I - \left(\frac{q_{\min}(G)}{n} - \frac{7}{50}\right)I\right)\left(\sum_{j=0}^{4} u_j U_j^{xy}\right)\right] \\
&= \mathbb{E}_{x,y}\left[\frac{A_{xy}}{n}(U^{xy}u)^{\mathsf{T}}\left(\frac{Q_G}{n} - \frac{7}{50}I - \frac{q_{\min}(G)}{n}I + \frac{7}{50}I\right)(U^{xy}u)\right] \\
&= \mathbb{E}_{x,y}\left[\frac{A_{xy}}{n}(U^{xy}u)^{\mathsf{T}}\left(\frac{Q_G - q_{\min}(G)I}{n}\right)(U^{xy}u)\right] \geq 0.
\end{aligned} \tag{33}$$

The last inequality is pointwise before averaging: (31) applied to $U^{xy}u$ gives

$$\begin{aligned}
&(U^{xy}u)^{\mathsf{T}}\left(\frac{Q_G - q_{\min}(G)I}{n}\right)(U^{xy}u) \\
&\quad = (U^{xy}u)^{\mathsf{T}}\left(\frac{Q_G}{n}\right)(U^{xy}u) - \frac{q_{\min}(G)}{n}\|U^{xy}u\|^2 \geq 0,
\end{aligned}$$

and $A_{xy}/n \geq 0$ preserves this inequality when multiplying and averaging. Because (33) holds for every $u \in \mathbb{R}^5$, the symmetric matrix $\mathsf{F} - (q_{\min}(G)/n - 7/50)\mathsf{D}$ is positive semidefinite.

For the matrix B in the hypotheses, linearity of the trace gives

$$\begin{aligned}&\operatorname{tr}(B\mathsf{F}) - \left(\frac{q_{\min}(G)}{n} - \frac{7}{50}\right)\operatorname{tr}(B\mathsf{D}) \\ &= \operatorname{tr}\left(B\left[\mathsf{F} - \left(\frac{q_{\min}(G)}{n} - \frac{7}{50}\right)\mathsf{D}\right]\right) \geq 0.\end{aligned}$$

The last inequality follows from Lemma 6, since both matrix factors are positive semidefinite. Rearranging gives

$$\left(\frac{q_{\min}(G)}{n} - \frac{7}{50}\right)\operatorname{tr}(B\mathsf{D}) \leq \operatorname{tr}(B\mathsf{F}). \tag{34}$$

Since B is positive definite, $B - \lambda_{\min}(B)I$ is positive semidefinite. Since D is also positive semidefinite, Lemma 6 and linearity of the trace give

$$\operatorname{tr}(B\mathsf{D}) - \lambda_{\min}(B)\operatorname{tr}(\mathsf{D}) = \operatorname{tr}\big((B - \lambda_{\min}(B)I)\mathsf{D}\big) \geq 0.$$

Thus, by (32) and $\lambda_{\min}(B) > 0$,

$$\operatorname{tr}(B\mathsf{D}) \geq \lambda_{\min}(B)\operatorname{tr}(\mathsf{D}) > 0.$$

By hypothesis, $\operatorname{tr}(B\mathsf{F}) \leq 0$. Combining this with (34) gives

$$\left(\frac{q_{\min}(G)}{n} - \frac{7}{50}\right)\operatorname{tr}(B\mathsf{D}) \leq \operatorname{tr}(B\mathsf{F}) \leq 0.$$

Dividing by the strictly positive quantity $\operatorname{tr}(B\mathsf{D})$ gives

$$\frac{q_{\min}(G)}{n} - \frac{7}{50} \leq \frac{\operatorname{tr}(B\mathsf{F})}{\operatorname{tr}(B\mathsf{D})} \leq 0, \qquad \text{hence} \qquad q_{\min}(G) \leq \frac{7}{50}n.$$

Since G was arbitrary, this proves Theorem 1. □

By Lemma 8, it remains to construct one real symmetric positive definite 5×5 matrix B, independent of G, for which $\operatorname{tr}(B\mathsf{F}) \leq 0$ for every finite simple triangle-free graph G with at least one edge.

To calculate F, we expand its entries into linear combinations of homomorphism densities. In the coordinate formulas below, write $U_i(z) = U_i^{xy}(z)$. The variables listed in each expectation subscript are sampled independently and uniformly from $V(G)$, with replacement. In averages over x, y, suppress only the superscript xy on each U_i, and first expand (30) as

$$\begin{aligned}\mathsf{F}_{ij} = \mathbb{E}_{x,y,z}\left[A_{xy}U_i(z)\frac{\deg_G(z)}{n}U_j(z)\right] \\ + \mathbb{E}_{x,y,z,t}\left[A_{xy}A_{zt}U_i(z)U_j(t)\right] - \frac{7}{50}\mathbb{E}_{x,y,z}\left[A_{xy}U_i(z)U_j(z)\right].\end{aligned}$$

The three underlying coordinate sums are

$$\begin{aligned}(U_i^{xy})^{\mathsf{T}}\operatorname{diag}_{v\in V}(\deg_G(v))U_j^{xy} &= \sum_z \deg_G(z)U_i(z)U_j(z), \\ (U_i^{xy})^{\mathsf{T}}AU_j^{xy} &= \sum_{z,t} A_{zt}U_i(z)U_j(t), \\ (U_i^{xy})^{\mathsf{T}}U_j^{xy} &= \sum_z U_i(z)U_j(z),\end{aligned}$$

and the factors $1/n$ in (30) give exactly the normalized averages in the first displayed expansion of F_{ij}. Substituting $\deg_G(z)/n = \mathbb{E}_t A_{zt}$ into the first term gives

$$\begin{aligned}\mathsf{F}_{ij} &= \mathbb{E}_{x,y,z,t}[A_{xy}A_{zt}U_i(z)U_j(z)] \\ &\quad + \mathbb{E}_{x,y,z,t}[A_{xy}A_{zt}U_i(z)U_j(t)] - \frac{7}{50}\mathbb{E}_{x,y,z}[A_{xy}U_i(z)U_j(z)].\end{aligned}$$

For fixed x, y, z, the first column vector has the expansion

$$U_0(z) = \frac{1}{n}\sum_s A_{zs}(A_{sx} - A_{sy}) - \frac{1}{50}(A_{zx} - A_{zy}) = \mathbb{E}_s[A_{zs}(A_{sx} - A_{sy})] - \frac{1}{50}(A_{zx} - A_{zy}).$$

Replacing each degree in (27) by its adjacency sum gives

$$\begin{aligned}U_1(z) &= \mathbb{E}_s[(A_{xs} - A_{ys})(A_{zx} + A_{zy})], \\ U_2(z) &= \mathbb{E}_s\left[\left(A_{zs} - \frac{A_{xs} + A_{ys}}{2}\right)(A_{zx} - A_{zy})\right].\end{aligned}$$

For fixed x, y, z, take the z-coordinate in (28) and use

$$(A^2)_{zt} = \sum_{s\in V} A_{zs}A_{st}, \qquad \sum_{t\in V} I_{zt}(A_{tx} - A_{ty}) = A_{zx} - A_{zy}.$$

Expanding the matrix entries, distributing the sum, and then normalizing gives

$$\begin{aligned}U_3(z) &= \sum_{t\in V}\left(\frac{A^2}{n^2} + \frac{3A}{50n} - \frac{1}{625}I\right)_{zt}(A_{tx} - A_{ty}) \\ &= \sum_{t\in V}\left(\frac{(A^2)_{zt}}{n^2} + \frac{3A_{zt}}{50n} - \frac{I_{zt}}{625}\right)(A_{tx} - A_{ty}) \\ &= \frac{1}{n^2}\sum_{t\in V}(A^2)_{zt}(A_{tx} - A_{ty}) + \frac{3}{50n}\sum_{t\in V}A_{zt}(A_{tx} - A_{ty}) \\ &\quad - \frac{1}{625}\sum_{t\in V} I_{zt}(A_{tx} - A_{ty}) \\ &= \frac{1}{n^2}\sum_{t\in V}\left(\sum_{s\in V}A_{zs}A_{st}\right)(A_{tx} - A_{ty}) + \frac{3}{50n}\sum_{t\in V}A_{zt}(A_{tx} - A_{ty}) \\ &\quad - \frac{1}{625}(A_{zx} - A_{zy}) \\ &= \frac{1}{n^2}\sum_{s,t\in V}A_{zs}A_{st}(A_{tx} - A_{ty}) + \frac{3}{50}\left(\frac{1}{n}\sum_{s\in V}A_{zs}(A_{sx} - A_{sy})\right) \\ &\quad - \frac{1}{625}(A_{zx} - A_{zy}) \\ &= \mathbb{E}_{s,t}[A_{zs}A_{st}(A_{tx} - A_{ty})] + \frac{3}{50}\mathbb{E}_s[A_{zs}(A_{sx} - A_{sy})] - \frac{1}{625}(A_{zx} - A_{zy}).\end{aligned}$$

Replacing the normalized sums by expectations in (29) gives

$$\begin{aligned}U_4(z) = \mathbb{E}_s\left[A_{zs} - \frac{A_{xs} + A_{ys}}{2}\right]\mathbb{E}_t\left[A_{zt}(A_{tx} - A_{ty}) - \frac{1}{50}(A_{zx} - A_{zy})\right] \\ + \frac{1}{4}(A_{zx} - A_{zy})\mathbb{E}_s[A_{xs} - A_{ys}]\,\mathbb{E}_t[A_{xt} - A_{yt}].\end{aligned}$$

Combining these independent averages gives the fully expanded fifth column vector

$$U_4(z) = \mathbb{E}_{s,t}\left[\left(A_{zs} - \frac{A_{xs}+A_{ys}}{2}\right)\left(A_{zt}(A_{tx}-A_{ty}) - \frac{1}{50}(A_{zx}-A_{zy})\right)\right] + \frac{1}{4}\mathbb{E}_{s,t}\left[(A_{xs}-A_{ys})(A_{xt}-A_{yt})(A_{zx}-A_{zy})\right].$$

Substitute the five expectation formulas above into the three terms of F_{ij} and distribute the finite products. By linearity of expectation, F_{ij} is a rational linear combination of expectation monomials of the form

$$\mathbb{E}_{v_1,\dots,v_s}\left[A_{v_{a_1}v_{b_1}}\cdots A_{v_{a_m}v_{b_m}}\right].$$

Here $v_1,\dots,v_s$ name the distinct positions occurring in the product; all are sampled independently and uniformly, and their sampled values may coincide. Inspecting the five column-vector formulas shows that each product obtained from the first two terms of F_{ij} uses at most eight positions, and each product obtained from its last term uses at most seven. Since $A_{ab} \in \{0,1\}$, repeated factors can be replaced by one factor; also $A_{ab} = A_{ba}$, so their order does not matter. Terms containing A_{aa} vanish because $A_{aa} = 0$, and terms containing $A_{ab}A_{bc}A_{ca}$ vanish because G is triangle-free. For each remaining term, form the simple graph J on $[s]$ with edge set

$$E(J) = \{\{a_r, b_r\} : 1 \le r \le m\}.$$

The term is exactly its homomorphism density:

$$\mathbb{E}_{v_1,\dots,v_s}\left[\prod_{\{a,b\}\in E(J)} A_{v_a v_b}\right] = \frac{1}{n^s}\sum_{(v_1,\dots,v_s)\in V(G)^s}\prod_{\{a,b\}\in E(J)} A_{v_a v_b} = t(J,G).$$

Thus, for $J \in \bigcup_{s=0}^{8}\mathcal{T}_s$, let $c_{ij,J} \in \mathbb{Q}$ be the sum of the rational coefficients of all terms whose graphs are isomorphic to J, with $c_{ij,J} = 0$ when there are no such terms. Then

$$\mathsf{F}_{ij} = \sum_{J\in\bigcup_{s=0}^{8}\mathcal{T}_s} c_{ij,J}\, t(J,G). \tag{35}$$

Exchanging i,j commutes the factors in the degree and norm terms, while exchanging z,t in the adjacency term and consistently relabeling its position variables gives a coefficient-preserving bijection of monomials, so $c_{ij,J} = c_{ji,J}$.

To express F as an average over $\mathcal{T}_8$, define, for each $H \in \mathcal{T}_8$, the rational symmetric 5×5 matrix F_H by

$$(F_H)_{ij} = \sum_J c_{ij,J}\, t_{\mathrm{inj}}(J,H) = \sum_J \frac{c_{ij,J}}{(8)_{|V(J)|}}\sum_{\phi:V(J)\hookrightarrow V(H)}\prod_{\{a,b\}\in E(J)}(A_H)_{\phi(a)\phi(b)}. \tag{36}$$

Applying Lemma 2 to every summand of (35) and interchanging finite sums gives

$$\mathsf{F}_{ij} = \sum_J c_{ij,J}\sum_{H\in\mathcal{T}_8} t_{\mathrm{inj}}(J,H)\Pr(H_X \cong H) = \sum_{H\in\mathcal{T}_8}(F_H)_{ij}\Pr(H_X\cong H), \qquad \mathsf{F} = \sum_{H\in\mathcal{T}_8} F_H \Pr(H_X\cong H), \tag{37}$$

where X is an independent uniform eight-tuple from V.

By Lemma 8, it remains to construct one real symmetric positive definite 5×5 matrix B, independent of G, and prove

$$\operatorname{tr}(B\mathsf{F}) \le 0 \qquad \text{for every finite triangle-free graph } G \text{ with at least one edge.}$$

Here F is the matrix defined from G in (30). By (37) and the linearity of the trace,

$$\operatorname{tr}(B\mathsf{F}) = \sum_{H \in \mathcal{T}_8} \Pr(H_X \cong H) \operatorname{tr}(BF_H).$$

Thus the only dependence on G in this expression is through the probabilities $\Pr(H_X \cong H)$. The matrices F_H are the fixed rational matrices in (36).

By Lemma 5, for each type σ, the matrix

$$\sum_{H \in \mathcal{T}_8} \Pr(H_X \cong H) C_H^\sigma \quad \text{is positive semidefinite.}$$

For each type $\sigma \in \bigcup_{k=0}^{6} \mathcal{T}_k$, we will choose a real symmetric positive semidefinite matrix M_σ, independent of G, whose rows and columns are indexed by $\mathcal{F}_\sigma$. We seek to express $\operatorname{tr}(B\mathsf{F})$ in the following form:

$$\operatorname{tr}(B\mathsf{F}) = -\sum_{\sigma} \operatorname{tr}\left(M_\sigma \sum_{H \in \mathcal{T}_8} \Pr(H_X \cong H) C_H^\sigma \right).$$

All sums over σ here and below range over $\bigcup_{k=0}^{6} \mathcal{T}_k$. The following lemma makes this reduction to the eight-vertex identities precise.

Lemma 9. *If there exist a real symmetric positive definite 5×5 matrix B and, for every type $\sigma \in \bigcup_{k=0}^{6} \mathcal{T}_k$, a real symmetric positive semidefinite matrix M_σ whose rows and columns are indexed by $\mathcal{F}_\sigma$, such that*

$$\operatorname{tr}(BF_H) = -\sum_{\sigma} \operatorname{tr}(M_\sigma C_H^\sigma) \qquad \text{for every } H \in \mathcal{T}_8,$$

then

$$\operatorname{tr}(B\mathsf{F}) \le 0$$

for every finite simple triangle-free graph G with $|V(G)| \ge 1$, where F is defined from G by (30). Consequently, Theorem 1 holds.

Proof. Fix an arbitrary finite simple triangle-free graph G with $|V(G)| \ge 1$, and let X be an independent uniform eight-tuple from $V(G)$. Using (37), the assumed identities, and linearity of the trace, we obtain

$$\begin{aligned} \operatorname{tr}(B\mathsf{F}) &= \sum_{H \in \mathcal{T}_8} \Pr(H_X \cong H) \operatorname{tr}(BF_H) \\ &= -\sum_{H \in \mathcal{T}_8} \Pr(H_X \cong H) \sum_{\sigma} \operatorname{tr}(M_\sigma C_H^\sigma) \\ &= -\sum_{\sigma} \operatorname{tr}\left(M_\sigma \sum_{H \in \mathcal{T}_8} \Pr(H_X \cong H) C_H^\sigma \right) \le 0. \end{aligned}$$

The third equality interchanges the finite sums and uses linearity of the trace. For each σ, the first matrix factor in the last line is positive semidefinite by hypothesis, and the second is positive

semidefinite by Lemma 5. Thus their trace product is nonnegative by Lemma 6, which gives the last inequality. Since G was arbitrary and B is positive definite, Lemma 8 proves Theorem 1. This finishes the proof. □

To find the matrices in Lemma 9, we used semidefinite programming with the symmetric matrices B, M_σ and a real scalar τ as unknowns, while F_H and C_H^σ were fixed:

$$\begin{aligned} \text{minimize} \quad & \tau \\ \text{subject to} \quad & B_{00} = 1, \qquad B - 10^{-8} I \succeq 0, \qquad \tau \geq 0, \\ & M_\sigma \succeq 0 \quad \left(\sigma \in \bigcup_{k=0}^{6} \mathcal{T}_k\right), \\ & -\tau \leq 10^6 \left(\operatorname{tr}(BF_H) + \sum_\sigma \operatorname{tr}(M_\sigma C_H^\sigma)\right) \leq \tau \quad (H \in \mathcal{T}_8). \end{aligned}$$

Here $S \succeq 0$ means that S is positive semidefinite, and the sizes of B and M_σ are those in Lemma 9. The normalization $B_{00} = 1$ excludes the zero solution, and the lower bound on B ensures positive definiteness. The objective minimizes the maximum absolute error in the 410 identities, scaled by 10^6; a feasible solution with $\tau = 0$ satisfies those identities exactly. The constant $7/50$ in F_H is fixed throughout this search. We then rationalized the numerical matrix factors and corrected the remaining errors by solving linear equations over the rationals, obtaining the exact integer data below, whose positivity and identities are checked in Lemma 10.

To specify explicitly the matrices obtained from the SDP after rational correction, we introduce the following integer data. To construct a positive definite B, take a positive integer q, an invertible integer 5×5 matrix L, and a symmetric integer 5×5 matrix X_*, and set

$$B = L(qI + X_*)L^{\mathsf{T}}.$$

We will choose them so that $qI + X_*$ is positive definite; then B is positive definite since L is invertible.

For each $\sigma \in \bigcup_{k=0}^{6} \mathcal{T}_k$, to construct a positive semidefinite M_σ, take a rectangular integer matrix V_σ whose rows are indexed by $P \in \mathcal{F}_\sigma$, and a symmetric integer matrix X_σ whose size equals the number of columns of V_σ. We will require $qI + X_\sigma$ to be positive definite as well. Then the square matrix

$$V_\sigma(qI + X_\sigma)V_\sigma^{\mathsf{T}}$$

has rows and columns indexed by the same flags as C_H^σ, and is positive semidefinite. For these constructions, Lemma 7 gives a direct way to check the required positive definiteness: it suffices to verify

$$q + (X_\tau)_{ii} > \sum_{j \neq i} |(X_\tau)_{ij}| \quad \text{for every row } i, \qquad \tau \in \{*\} \cup \bigcup_{k=0}^{6} \mathcal{T}_k.$$

The trace of each such product means the explicit finite sum

$$\operatorname{tr}\left(V_\sigma(qI + X_\sigma)V_\sigma^{\mathsf{T}} C_H^\sigma\right) = \sum_{P,P' \in \mathcal{F}_\sigma} [V_\sigma(qI + X_\sigma)V_\sigma^{\mathsf{T}}]_{PP'} C_H^\sigma(P, P').$$

For positive integers a_σ, we choose

$$M_\sigma = \frac{2^{80}}{8!\, a_\sigma^2} V_\sigma(qI + X_\sigma)V_\sigma^{\mathsf{T}}.$$

This matrix is positive semidefinite because $2^{80}/(8!a_\sigma^2) > 0$ and $V_\sigma(qI+X_\sigma)V_\sigma^{\mathsf{T}}$ is positive semidefinite. Thus the fixed identities to verify become

$$\operatorname{tr}(BF_H) = -\sum_\sigma \frac{2^{80}}{8!\,a_\sigma^2} \operatorname{tr}\left(V_\sigma(qI+X_\sigma)V_\sigma^{\mathsf{T}} C_H^\sigma\right) \qquad (H \in \mathcal{T}_8).$$

The coefficients $2^{80}/a_\sigma^2$ are fixed positive rational numbers in the representation; their numerical values will be specified below. By Lemma 9, these identities and the stated positivity suffice to prove Theorem 1. It remains to specify $L, q, X_*, X_\sigma, V_\sigma$ explicitly and check the following conditions:

(i) The integer matrix L is invertible.

(ii) The matrix $qI + X_*$ is positive definite. By Lemma 7, it suffices to check

$$q + (X_*)_{ii} - \sum_{j\neq i} |(X_*)_{ij}| > 0 \qquad (i = 0, \dots, 4).$$

(iii) For every type σ, the matrix $qI + X_\sigma$ is positive definite. By Lemma 7, it suffices to check

$$q + (X_\sigma)_{ii} - \sum_{j\neq i} |(X_\sigma)_{ij}| > 0 \qquad \text{in every row } i.$$

(iv) For a fixed positive integer Δ, the following identity holds for every $H \in \mathcal{T}_8$:

$$\begin{aligned} 0 = {} & \frac{\Delta}{2^{80}} \operatorname{tr}\left(L(qI+X_*)L^{\mathsf{T}} F_H\right) \\ & + \sum_\sigma \frac{\Delta}{40320 a_\sigma^2} \operatorname{tr}\left(V_\sigma(qI+X_\sigma)V_\sigma^{\mathsf{T}} C_H^\sigma\right). \end{aligned}$$

We now briefly introduce how we compute the required coefficients and matrices, specify the fixed integer data, and verify these conditions.[1]

Step 1. Computing $c_{ij,J}$. The program `programs/01_expand_F.py` implements the five expectation formulas for $U_0(z), \dots, U_4(z)$ displayed above and substitutes them into F_{ij}. Each term is stored as its rational coefficient and a list of adjacency factors, with a distinct name for each distinct summation position. In the complete expansion of F_{ij}, every such position is averaged independently and uniformly; this convention is not stored as additional data. Addition, scalar multiplication and multiplication first expand the entire expression by the distributive law. The program then removes terms containing a loop or triangle, removes repeated adjacency factors, and adds the coefficients of isomorphic graphs. The resulting coefficients are precisely those in (35). The output `programs/results/01/c_ij_J.pkl` is a dictionary with keys (i,j), $0 \le i,j \le 4$; its value at (i,j) is the dictionary $\{J \mapsto c_{ij,J}\}$. A graph is stored as a pair `(vertices, edges)` of tuples: the first tuple lists its vertices, and the second lists its edges as pairs (a,b) with $a < b$, representing the unordered edges $\{a,b\}$. The stored vertex set is $[s]$ for a graph of order s. All coefficients are exact rational numbers, stored using `Fraction`.

[1] The programs, data and manuscript are in the GitHub repository `Diudiutse/triangle-free-signless-laplacian`. All file paths below are relative to its root.

Step 2. Generating $\mathcal{T}_8$. The program programs/02_generate_tau8.py starts with the empty graph on no vertices and adds one vertex at a time. For every retained graph on $[s]$, it considers every independent set $S \subseteq [s]$ and adds vertex $s+1$ with neighborhood exactly S. It then retains one representative of each ordinary isomorphism class, using NetworkX for the isomorphism test. The new graph is triangle-free because any new triangle would contain $s+1$ and two adjacent vertices in S. Conversely, deleting a vertex from any triangle-free graph leaves a triangle-free graph in which that vertex's former neighborhood is independent. Thus induction on the number of vertices proves that the enumeration is complete. The numbers of representatives at orders $0, \dots, 8$ are

$$1, 1, 2, 3, 7, 14, 38, 107, 410.$$

The output programs/results/02/tau_8.pkl is the list of the 410 representatives $H \in \mathcal{T}_8$, in the generated order.

Step 3. Computing F_H. The program programs/03_compute_F_H.py loads the coefficient dictionary from programs/results/01/c_ij_J.pkl and the graph list from programs/results/02/tau_8.pkl. Its function `tinj(J,H)` enumerates all injections $\phi : V(J) \hookrightarrow V(H)$, counts those preserving every edge of J, and divides by the total number of injections. It therefore returns $t_{\mathrm{inj}}(J, H)$ as an exact rational number. For each H and each (i, j), the program evaluates exactly the sum in (36):

$$(F_H)_{ij} = \sum_J c_{ij,J} t_{\mathrm{inj}}(J, H).$$

The output programs/results/03/F_H.pkl is a dictionary whose key is H and whose value is the dictionary $\{(i, j) \mapsto (F_H)_{ij}\}$. It contains 410 matrices, each with 25 rational entries.

Step 4. Generating the types σ. The program programs/04_generate_sigma.py calls the graph-generation function from programs/02_generate_tau8.py for each order $k = 0, \dots, 6$. It lists the resulting representatives in increasing order of k and saves them in programs/results/04/sigma_list.pkl. There are $1 + 1 + 2 + 3 + 7 + 14 + 38 = 66$ types. For a retained representative on $[k]$, these vertex numbers become the fixed root labels in the subsequent flag computation.

Step 5. Generating $\mathcal{F}_\sigma$. The program programs/05_generate_flags.py reads the type list from step 4. For each σ of order k, it starts with σ and successively adds vertices $k+1, \dots, \ell$, where $\ell = \lfloor (8+k)/2 \rfloor$, again considering every independent neighborhood. At each order, it identifies two graphs only when an isomorphism fixes $1, \dots, k$ individually. The function `labeled_graphs_are_isomorphic` tests this condition by enumerating all permutations of the remaining vertices and checking whether one of them preserves every adjacency and nonadjacency. Deleting an unlabeled vertex gives the same induction as in step 2, now with all root labels fixed, so every σ-flag of order ℓ is represented. The output programs/results/05/F_sigma.pkl is the dictionary $\{\sigma \mapsto$ the ordered list of flags in $\mathcal{F}_\sigma\}$. This list fixes the row and column order of C_H^σ and the row order of V_σ.

Step 6. Computing C_H^σ. The program programs/06_compute_C.py reads the graph list from step 2 and the flag lists from step 5. For each (σ, H), the function `compute_C_matrix` computes the entire matrix in (18). It enumerates all ordered choices of k distinct roots in H inducing exactly the labeled type σ. For each such root map θ, it enumerates pairs of subsets satisfying

$$S, T \subseteq V(H) \setminus \theta([k]), \qquad |S| = |T| = \ell - k, \qquad S \cap T = \varnothing.$$

It finds the flag representatives P, P' for which

$$(H[\theta([k]) \cup S], \theta) \cong_\sigma P, \qquad (H[\theta([k]) \cup T], \theta) \cong_\sigma P',$$

and adds

$$((\ell - k)!)^2 (8 - 2\ell + k)!$$

to the entry indexed by (P, P'). The two factors $(\ell - k)!$ count the orderings within S and T, and the last factorial counts the orderings of the unused vertices. The root order has already been fixed, so this counts every bijection in (18) exactly once. The output `programs/results/06/C.pkl` is the dictionary $\{(\sigma, H) \mapsto C_H^\sigma\}$, containing $66 \cdot 410 = 27060$ complete integer matrices in the flag orders from step 5.

Step 7. Specifying the fixed integers. After these six computations, the remaining inputs are the fixed integers L, q, Δ, X_* and $V_\sigma, X_\sigma, a_\sigma$. The file `programs/spectral_data.py` supplies L, q, Δ, X_*, with $q, \Delta > 0$, and

$$L = \begin{pmatrix} 1099511627776 & 0 & 0 & 0 & 0 \\ -232335715041 & 58226488264 & 0 & 0 & 0 \\ 646804735047 & 36701051459 & 62858792171 & 0 & 0 \\ -3529332685417 & -399541705998 & 9914953481 & 414703530043 & 0 \\ -47136593763 & -33884256159 & 53733927090 & -22938839431 & 195039391412 \end{pmatrix}.$$

It is lower triangular with positive diagonal entries, so

$$\det L = 1099511627776 \cdot 58226488264 \cdot 62858792171 \cdot 414703530043 \cdot 195039391412 > 0.$$

This verifies condition (i).

The file `programs/sigma_data.py` supplies a dictionary `sigma_data`. Its key is a type σ from step 4; the value is a dictionary with exactly three entries: `V_sigma` stores V_σ, `X_sigma` stores X_σ, and `a_sigma` stores the positive integer a_σ. The rows of V_σ follow the flag list from step 5, and the indices of X_σ follow the columns of V_σ. The symmetric matrices X_* and X_σ are specified in these files by upper-triangular triples (i, j, c), with $i \le j$. For $\tau = *$ or a type σ, these triples determine the full matrix by

$$(X_\tau)_{ij} = (X_\tau)_{ji} = \begin{cases} c, & \text{if the triple } (i,j,c) \text{ is listed for } X_\tau, \\ 0, & \text{if no triple is listed for } X_\tau \text{ at } (i,j), \end{cases} \qquad (i \le j). \tag{38}$$

Matrix indices in the programs start at zero. Importing the two files produces the complete integer matrices with these entries.

Step 8. Verifying the remaining conditions. The last two programs check conditions (ii), (iii) and (iv) using these inputs. Their exact conclusions are as follows.

Lemma 10. *For the fixed integer data just specified, every $\tau \in \{*\} \cup \bigcup_{k=0}^{6} \mathcal{T}_k$ satisfies*

$$q + (X_\tau)_{ii} - \sum_{j \ne i} |(X_\tau)_{ij}| > 0 \qquad \text{in every row } i, \tag{39}$$

and every $H \in \mathcal{T}_8$ satisfies

$$0 = \frac{\Delta}{2^{80}} \operatorname{tr}\left(L(qI + X_*)L^{\mathsf{T}} F_H\right) + \sum_\sigma \frac{\Delta}{40320 a_\sigma^2} \operatorname{tr}\left(V_\sigma (qI + X_\sigma) V_\sigma^{\mathsf{T}} C_H^\sigma\right). \tag{40}$$

Proof by exact finite verification. First, `programs/verify_39.py` imports q, X_* from `programs/spectral_data.py` and the dictionary `sigma_data` from `programs/sigma_data.py`. Its function `satisfies_39(X)` checks

$$q + X_{ii} > \sum_{j \neq i} |X_{ij}|$$

in every row of the input matrix X. The program applies this function first to X_*, and then to each X_σ in the dictionary. It stops and reports the corresponding matrix if any inequality fails. All 1650 rows of the 67 matrices pass this check, proving (39). Lemma 7 then verifies conditions (ii) and (iii).

Next, `programs/verify_40.py` imports L, q, Δ, X_* from `programs/spectral_data.py` and all $V_\sigma, X_\sigma, a_\sigma$ from `programs/sigma_data.py`. It reads the four outputs

`programs/results/02/tau_8.pkl`, `programs/results/03/F_H.pkl`,
`programs/results/05/F_sigma.pkl`, `programs/results/06/C.pkl`.

Before checking the identity, it checks that the dictionary keys cover exactly the required graphs and types, that all C_H^σ have the sizes determined by the flag lists, that the row counts of V_σ agree with those lists, and that $a_\sigma > 0$ and $\Delta/(40320a_\sigma^2) \in \mathbb{Z}$ for every type.

It forms $B = L(qI + X_*)L^\mathsf{T}$ once, and then loops over every H in the graph list from step 2. For that H, it reads F_H and the matching matrices C_H^σ and evaluates the entire left-hand side of (40). The first trace is computed as $\operatorname{tr}(BF_H)$. For each term in the sum over σ, the program uses

$$\begin{aligned} \operatorname{tr}\left(V_\sigma(qI + X_\sigma)V_\sigma^\mathsf{T} C_H^\sigma\right) &= \operatorname{tr}\left((qI + X_\sigma)(V_\sigma^\mathsf{T} C_H^\sigma V_\sigma)\right) \\ &= \sum_{i,j} (qI + X_\sigma)_{ji} (V_\sigma^\mathsf{T} C_H^\sigma V_\sigma)_{ij}. \end{aligned} \tag{41}$$

The first equality uses cyclicity of the trace, and the second is its entrywise definition. The code implements the last sum by transposing the first matrix, multiplying corresponding entries of the two matrices, and summing all entries. All matrix operations use NumPy arrays with `dtype=object`, so the entries remain Python integers or exact rational `Fraction` values. The program stops if the value for any H is nonzero. All 410 values are exactly zero, proving (40) and condition (iv). The output `programs/results/verify_40.json` records these 410 zero values. This finishes the verification of Lemma 10. □

Proof of Theorem 1: The result follows from Lemmas 8, 9 and 10. □

Statement on AI use

The authors used GPT-6.1 Sol in the preparation of this work, including the development of proof arguments, the checking of intermediate steps, the generation and execution of semidefinite programming (SDP) code and code for verifying inequalities and equalities, and the drafting and revision of the manuscript and its LaTeX source. All mathematical results, arguments, and proofs were independently reviewed and verified by the authors, who take full responsibility for the accuracy and content of this work.

References

[1] J. Balogh, F. C. Clemen, B. Lidický, S. Norin and J. Volec, *The spectrum of triangle-free graphs*, SIAM J. Discrete Math. **37** (2023), 1173–1179. doi:10.1137/22M150767X.

[2] S. Boyd and L. Vandenberghe, *Convex Optimization*, Cambridge University Press, Cambridge, 2004. doi:10.1017/CBO9780511804441.

[3] S. Brandt, *The local density of triangle-free graphs*, Discrete Math. **183** (1998), 17–25. doi:10.1016/S0012-365X(97)00074-5.

[4] L. de Lima, V. Nikiforov and C. Oliveira, *The clique number and the smallest Q-eigenvalue of graphs*, Discrete Math. **339** (2016), 1744–1752. doi:10.1016/j.disc.2016.02.002.

[5] L. Lovász, *Large Networks and Graph Limits*, American Mathematical Society Colloquium Publications, vol. 60, American Mathematical Society, Providence, RI, 2012. doi:10.1090/coll/060.

[6] A. A. Razborov, *Flag algebras*, J. Symbolic Logic **72** (2007), 1239–1282. doi:10.2178/jsl/1203350785.

[7] R. S. Varga, *Geršgorin and His Circles*, Springer Series in Computational Mathematics, vol. 36, Springer, Berlin, 2004. doi:10.1007/978-3-642-17798-9.

[8] Q. Zhou, *The least signless Laplacian eigenvalue of* $\{C_3, C_5\}$*-free graphs*, preprint (2026), arXiv:2610.00875.